

From 2D Pose Estimation to 3D Human Recovery

A Literature Review of OpenPose, HRNet, and HMR

Yiqiao Liu
yl14487@nyu.edu

Taijia Liang
tl4549@nyu.edu

I. INTRODUCTION

Human pose estimation has long been a core problem in computer vision, with applications in action recognition, behavior analysis, surveillance, and game production. It aims to extract and recover the structural information of the human body from images or videos. However, due to factors such as loss of image detail, occlusion, multi-person spatial interactions, viewpoint changes, and 3D depth ambiguity, this problem remains highly challenging [1]–[3].

This review examines three representative works—OpenPose, High-Resolution Network (HRNet), and Human Mesh Recovery (HMR)—through two perspectives: the number of people addressed in 2D pose estimation and the form of the final output representation. Through comparing these methods, this paper asks how human pose estimation has evolved from understanding 2D poses to recovering 3D human bodies, and what roles representation design, network architecture, and supervision strategy have played in that progression.

II. 2D HUMAN POSE ESTIMATION: FROM PRECISE SINGLE-PERSON LOCALIZATION TO MULTI-PERSON PARSING

In single-person 2D pose estimation, traditional methods typically extract features by downsampling the input image, perform pose estimation on the reduced-resolution representation with a Convolutional Neural Network (CNN), and then restore spatial information through upsampling. However, Sun et al. argue that high-resolution representations contain substantial spatial information, much of which is weakened when resolution is repeatedly reduced and later recovered, leading to less accurate spatial localization [2].

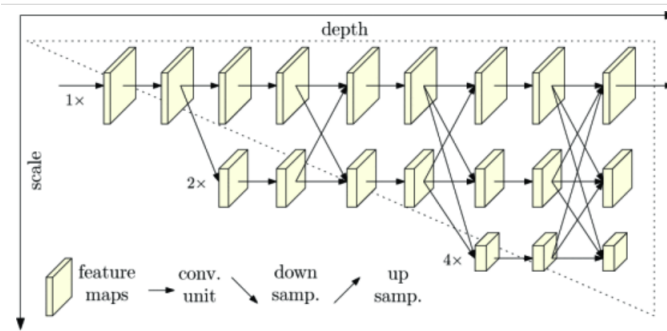


Fig. 1: Parallel multi-resolution subnetworks and repeated multi-scale fusion in HRNet [2].

Sun et al. proposed High-Resolution Network (HRNet), which maintains high-resolution representations throughout the network [2]. As shown in Figure 1, the core of HRNet combines parallel multi-resolution subnetworks with repeated multi-scale fusion. The parallel subnetworks are organized into four stages. The first stage contains only a high-resolution branch, and each subsequent stage retains the previous branches while adding a lower-resolution branch. This design enriches high-resolution features with stronger semantic information. Meanwhile, multi-scale fusion exchanges information across branches of different resolutions to generate aligned feature maps, and repeated multi-scale fusion performs this exchange several times. Through repeated downsampling and upsampling between branches, HRNet produces representations with both stronger semantic content and finer spatial detail [2]. Overall, HRNet improves the precise localization of body joints in 2D.

Although precise single-person localization provides an important foundation for 2D pose estimation, directly extending single-person methods to multi-person scenes introduces additional challenges. Cao et al. point out that multi-person estimation involves uncertainty caused by the unknown number of people in an image, spatial interference from interactions and overlap, and rapidly increasing parsing complexity as the number of people grows [1]. To address these issues, they propose Part Affinity Fields (PAFs), which model the association between human body parts through a set of 2D vector fields, together with confidence maps that indicate the locations of individual body parts [1]. Compared with traditional top-down methods that reuse single-person pose estimation and prior bottom-up methods, this design enables more efficient identification and reconstruction of unstructured pairwise relationships among body parts in multi-person settings [1].

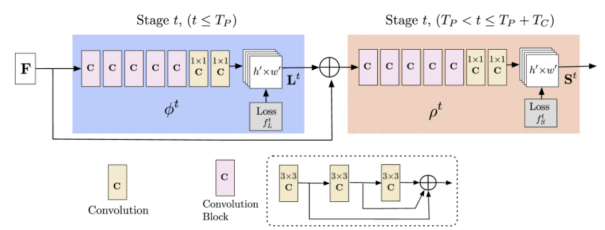


Fig. 2: Multi-stage prediction network used in OpenPose [1].

The OpenPose system takes a $w \times h$ color image as input and applies a multi-stage prediction network based on Convolutional

Neural Networks (CNNs), as shown in Figure 2 [1]. The network iteratively refines both PAFs and confidence maps. In the design, a 7×7 convolution is replaced by three consecutive 3×3 convolutions, which reduces the operation count from $2 \times 7^2 - 1 = 97$ to 51 [1]. A spatially weighted loss is further used to handle the fact that not all individuals can be fully labeled in crowded scenes. For multi-person parsing, Cao et al. simplify the global matching problem with a spanning-tree skeleton and bipartite matching over multiple limbs, and then solve the final association through greedy inference [1]. Generally, OpenPose shifts the focus from precise single-person localization to scalable multi-person association.

III. FROM 2D POSE REPRESENTATION TO 3D HUMAN RECOVERY

The two methods discussed above produce different forms of 2D output. HRNet represents the locations of body joints as keypoint probability heatmaps, whereas OpenPose reconstructs posture through confidence maps, Part Affinity Fields, and a non-fully-connected tree structure for multi-person parsing. However, from a practical perspective, using 3D human recovery as the final output is more meaningful, because it provides a more complete description of human pose, body orientation, and shape.

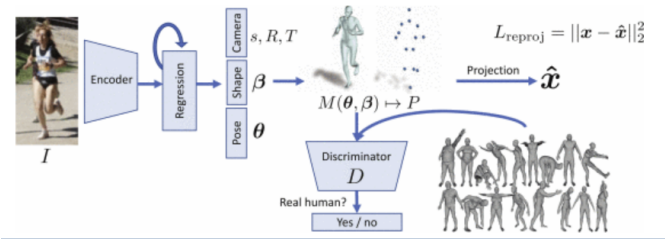


Fig. 3: HMR directly regresses SMPL parameters from a single RGB image to recover a 3D human mesh [3].

Against this background, methods focusing on 3D joints have emerged. Nevertheless, because such approaches still rely on mapping 2D joints to 3D parameters, some image information is inevitably lost. Kanazawa et al. proposed an end-to-end method called Human Mesh Recovery (HMR), which directly predicts 3D model parameters from image pixels [3]. HMR regresses the pose, shape, and camera parameters of the parameterized human body model SMPL from a single RGB image, thereby generating a complete 3D human mesh representation, as shown in Figure 3. Its core idea is twofold. On the one hand, a 2D reprojection loss ensures that the predicted 3D joints, when projected back to the image plane, align with the 2D annotations. On the other hand, an adversarial prior constrains the recovered results to remain within reasonable distributions of human poses and body shapes. This design addresses two major difficulties: the lack of reliable 3D annotations in real-world scenes and the inaccuracy of 2D-to-3D conversion caused by depth ambiguity [3].

IV. CONNECTIONS AND LIMITATIONS

When establishing connections among the three methods, it is necessary to revisit why Kanazawa et al. rejected traditional approaches that focus only on 3D joints. HMR does not negate 2D pose estimation itself; rather, it rejects the two-stage process that first compresses an image into sparse 2D keypoints and then relies only on these keypoints for 3D reconstruction [3]. Once image information is simplified to a small number of 2D joint positions, substantial cues about body contour, orientation, local occlusion, and shape detail are lost [3]. In contrast, although HRNet and OpenPose still output 2D representations, they improve the accuracy of single-person keypoint localization [2] and body-part association in multi-person scenes [1], respectively, thereby providing stronger 2D structural evidence for subsequent 3D reconstruction. In this sense, the three methods are not mutually exclusive, but can instead form a progressive technical pathway from 2D pose analysis to 3D human recovery.

In addition, each method has its own limitations:

- HRNet: Because it maintains high-resolution representations throughout the network, its computational cost is relatively high.
- OpenPose: Its output remains sensitive to complex occlusion and crowded scenes. When a person is too small or lacks detail, the prediction may deviate substantially.
- HMR: Its output is still affected by ambiguity, model assumptions, and supervision constraints.

V. FUTURE DIRECTIONS AND CONCLUSION

From HRNet to OpenPose, and then to HMR, these methods demonstrate the possibilities of human pose estimation in computer vision, from more accurate single-person 2D pose representation to complex multi-person 2D pose estimation and further toward complete 3D human body recovery. Together, these three methods indicate that progress in human pose estimation depends not only on more complex network architectures, but also on richer representations and more effective supervision strategies. At the same time, they expose several remaining limitations, which motivate future research directions. For example, performing 3D human body recovery in multi-person scenarios remains challenging, especially in the presence of complex spatial interactions and occlusion, and achieving stable and accurate 3D recovery in such settings still requires further exploration.

REFERENCES

- [1] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, "OpenPose: Realtime multi-person 2D pose estimation using part affinity fields," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 1, pp. 172–186, 2019.
- [2] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 5693–5703.
- [3] A. Kanazawa, M. J. Black, D. W. Jacobs, and J. Malik, "End-to-end recovery of human shape and pose," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7122–7131.